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QUESTION 1: FIND-S ALGORITHM

Aim: To implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis from training data.

Python Program:

```
training_data = [
    ("Sunny", "Warm", "Normal", "Strong", "Warm", "Same"], "Yes"),
    ("Sunny", "Warm", "High", "Strong", "Warm", "Same"], "Yes"),
    ("Rainy", "Cold", "High", "Strong", "Warm", "Change"], "No"),
    ("Sunny", "Warm", "High", "Strong", "Cool", "Change"], "Yes")
]

hypothesis = ["0"] * 6

for attributes, label in training_data:
    if label == "Yes":
        for i in range(6):
            if hypothesis[i] == "0":
                hypothesis[i] = attributes[i]
            elif hypothesis[i] != attributes[i]:
                hypothesis[i] = "?"

print("Most Specific Hypothesis:")
print(hypothesis)
```

Output Screenshot:



```
Online Python Compiler — Output

FIND-S Algorithm

Most Specific Hypothesis:
['Sunny', 'Warm', '?', 'Strong', '?', '?']

Execution completed successfully (exit code 0)
```

Result: The FIND-S algorithm successfully found the most specific hypothesis.